Mid-Semester Review: Speaker Notes & Talking Points

Mondrian Conformal Prediction for Uncertainty Quantification in ER Discrete-Event Simulation Surrogates · Course 23AID201

Speaking order: each person presents a contiguous block, not alternating - Rithvik (Slides 1-4, intro) → Venu (5-8, approach & methodology) → Vipin (9-11, data & core finding) → Harshith (12-14, detail & wrap-up)

Slide 1: Title Slide

Speaker: Rithvik | *On screen: Mondrian Conformal Prediction for Uncertainty Quantification in ER Discrete-Event Simulation Surrogates*

- Welcome everyone - we're presenting our mid-semester review for course 23AID201.
- Our project applies conformal prediction, specifically Mondrian conformal prediction, to a surrogate model trained on an emergency-department discrete-event simulation.
- Quick intros: I'm Rithvik, and with me are Venu, Vipin, and Harshith - I'll take the introduction, then each of them will take over in turn for one section apiece, so you'll hear from all four of us in sequence.
- We're at roughly 80 percent of the project now - what's built, validated, and checked so far, and what's left for end-sem.

Slide 2: Contents

Speaker: Rithvik | *On screen: Roadmap for this presentation*

- Here's the roadmap for the next few minutes.
- We'll start with the problem we're solving and the specific research gap we're addressing.
- Then two short concept slides - one explaining why a marginal coverage number can be misleading, and one showing the Mondrian partition idea - before we get into our actual bridging approach.
- After that: methodology, the real dataset we calibrated on, what's concretely done so far, our core finding, the simulation in more detail, a snapshot of our literature review, and finally what's left for end-sem.

Slide 3: Problem Statement

Speaker: Rithvik | *On screen: Why point predictions from a surrogate aren't enough*

- Surrogate models are fast, learned stand-ins for expensive simulations, and they're increasingly used to guide real decisions in systems like ER staffing or call-center queueing.
- But a single point prediction isn't enough for a decision like 'will adding two more doctors actually bring the wait time down' - you need a calibrated interval, not just one number.

- Two standard ways to get that interval: Gaussian Processes, which are theoretically solid but expensive to train and rely on distributional assumptions rather than a hard guarantee; and conformal prediction, which is distribution-free and comes with a finite-sample guarantee, but has only been validated for surrogate models in a narrow set of domains so far.
- That's the central question our project is built around: does conformal prediction still hold up when the surrogate is trained on a discrete-event queueing simulation, instead of the physics domains it's already been tested on?

Slide 4: Research Gap

Speaker: Rithvik | *On screen: What Gopakumar et al. (2026) leave open*

- Our base paper, Gopakumar et al. 2026, validates conformal prediction for surrogate-model uncertainty quantification, but only in physics domains - PDEs, magnetohydrodynamics, weather, fusion plasma modeling.
- They explicitly flag two open limitations of their own results: first, the guarantee they validate is marginal - averaged over the whole input space, with no guarantee it holds within any specific subgroup or condition. Second, their validation assumes calibration and test data are exchangeable, and they never test what happens under distribution shift.
- Neither limitation has been tested outside physics simulation. A discrete-event, queueing-driven system is structurally different - discrete stochastic arrivals and departures, shared-resource contention, priority queues - not a continuous PDE field.
- That's exactly the gap our project sits in, and it's the reason this domain is a genuinely new test case, not just a repeat of an already-answered question.
- I'll hand it over to Venu now, who'll walk through our actual approach, starting with a quick concept slide.

Slide 5: Why Marginal Coverage Isn't Enough (concept)

Speaker: Venu | *On screen: An illustrative example - not our own results*

- This chart is illustrative - a generic example, not our project's own numbers. It's here to make one specific concept concrete before we describe our approach; our own real version of this exact idea comes later, on Slide 11.
- Imagine four categories of a system. If you average coverage across all four, you get about 90.5 percent - it looks like your 90 percent target is comfortably met, and the red line marks exactly where that 90 percent target sits relative to each bar.
- But look closer: Category D alone sits at 68 percent, well below target. A decision-maker who only sees the marginal, averaged number would never know this specific category is unreliable.
- This is exactly the gap Mondrian conformal prediction is designed to close, by calibrating separately within each category instead of pooling everything into one marginal number.

Slide 6: Our Bridging Approach

Speaker: Venu | *On screen: Applying Mondrian CP, testing both limitations directly*

- Our core idea: apply conformal prediction, specifically Mondrian CP, to an ER queueing surrogate, and test directly whether Gopakumar et al.'s two limitations actually show up in this domain.
- On the left: addressing marginal coverage. Mondrian CP partitions calibration by category - staffing tercile crossed with arrival-rate tercile - instead of lumping every scenario into one marginal guarantee. That's done: Slide 11 confirms the worst category's coverage recovers from 68.2 to 90.9 percent, statistically significant across 30 repeats.
- On the right: addressing exchangeability, with an out-of-distribution 'surge day' stress test outside the normal training range - this is the main piece of work still remaining, the core of our final ~20 percent.
- That stress test is where we'll get to see whether standard CP and Mondrian CP hold up, or break down, once exchangeability is genuinely violated.

Slide 7: The Mondrian Partition (concept)

Speaker: Venu | *On screen: How the 9-category grid works*

- This is the actual partition structure Mondrian CP uses - not a future plan, this is the real structure behind the result on Slide 11.
- Every scenario in our simulation has two covariates: staffing capacity and arrival-rate multiplier. We split each into three terciles - low, medium, high - giving a 3-by-3 grid of 9 categories.
- Every simulated scenario falls into exactly one of these 9 cells, based on its own staffing level and arrival-rate multiplier - nothing is shared across cells. Mondrian CP calibrates a separate quantile within each one, instead of one pooled quantile across all 9.
- The highlighted category is comparatively low-stress - medium staffing, low arrival rate. The opposite corner - low staffing, high arrival rate - is the real category, in our own data, where Slide 11's 68.2-to-90.9-percent recovery actually happens.

Slide 8: Methodology Overview

Speaker: Venu | *On screen: The three-stage pipeline and how each stage is checked*

- Our pipeline has three stages, shown left to right: a discrete-event simulation built in SimPy and calibrated on real ED data; a surrogate model - gradient boosting - trained on the DES's scenario sweep; and uncertainty quantification, comparing a GP baseline against standard CP and Mondrian CP.
- We don't just chain these stages together and hope for the best - each one is checked against the previous one before we move on.
- The DES's output is checked against real aggregated ED statistics. Surrogate accuracy is checked against DES outputs held out from training, using MAE, RMSE, and R-squared.
- And critically, all three UQ methods use the identical fixed test split and the identical target coverage of 90 percent, so GP, standard CP, and Mondrian CP end up directly comparable. All three stages are complete now - the Uncertainty Quantification stage's headline result is on Slide 11.

- Over to Vipin now, who'll walk through the real dataset and our actual results.

Slide 9: Real-World Data Used

Speaker: Vipin | *On screen: The Kaggle dataset and what's real vs. literature-calibrated*

- We calibrate on the Hospital Triage and Patient History Data from Kaggle - real retrospective data from the Yale New Haven Health System, covering March 2014 through July 2017.
- It's a large dataset: 560,486 ED visits total, 972 variables per visit, across three EDs - one academic and two community sites. We calibrate on Department A only, the largest, academic site: 322,283 visits, about 258 per day.
- We're deliberate about keeping real data and literature-sourced values separate, not blending them silently. From the real data: arrival pattern by 4-hour bin, day of week and month, ESI acuity mix, and department-level daily visit rate.
- From literature: service and treatment time per ESI level, modeled as log-normal - the dataset has no discharge timestamp, and we checked across all 972 columns to be sure before falling back to literature values.

Slide 10: Work Completed So Far (~80%)

Speaker: Vipin | *On screen: DES validation, surrogate accuracy, and the GP baseline*

- This slide is our concrete progress checkpoint - we're at roughly 80 percent of the project now.
- First, DES validation: across 200 simulated days, our simulation's mean visits per day is 235.1, against a real value of 258.2 - a 91 percent match, which we consider a strong validation given how much of the calibration is real-data-driven.
- Second, surrogate accuracy: our gradient-boosting surrogate hits an R-squared of 0.929 on total patient count, and even the hardest target - the 95th-percentile wait time - reaches 0.647.
- Third, the GP baseline: at a 90 percent target, it undercovers on three of four targets, landing around 87.7 to 88.8 percent. That's not surprising - GPs rely on distributional assumptions rather than a finite-sample guarantee. Standard CP and Mondrian CP both close most of that gap on average - the next slide shows the real, more important result underneath that average.

Slide 11: Core Finding: Mondrian CP Closes the Coverage Gap

Speaker: Vipin | *On screen: The real result: 68.2% -> 90.9% in the worst category*

- This is our central finding, and it's done, not projected.
- On average - marginal coverage - all three methods land close to the 90 percent target: GP baseline around 89 percent, standard CP and Mondrian CP both just above 90. On its own, this table would look like a minor, almost unremarkable improvement.
- But checked within each of the 9 categories rather than pooled, standard CP's coverage in the single worst category - understaffed with a high arrival rate - collapses to 68.2 percent, well under target. That failure is completely invisible in the marginal average above.

- Mondrian CP, calibrating separately per category, restores that same category to 90.9 percent coverage, without sacrificing the marginal guarantee. And this isn't a one-off lucky split - it's confirmed statistically significant across 30 independent repeats, paired t-test, p less than 0.001.
- I'll pass it to Harshith now to close out with the simulation in more detail, our literature review, and what's left to do.

Slide 12: The Discrete-Event Simulation

Speaker: Harshith | *On screen: What the DES actually models, and how it's validated*

- This chart shows the real arrival volume by 4-hour bin that our simulation is calibrated on - you can see the clear peak in the 11-to-14 window, which matches typical ED demand patterns.
- Our simulation is a SimPy discrete-event model of a single ED, calibrated directly on this real arrival pattern plus the real ESI acuity mix - not a generic textbook queueing example.
- It uses a priority-resource pool, where higher-acuity patients - lower ESI numbers - get priority for capacity, matching how real triage actually works.
- And as we mentioned on Slide 10, it's validated against real aggregated daily volume at a 91 percent match across 200 simulated days - this simulation is the foundation everything downstream, the surrogate and the uncertainty quantification, is built on.

Slide 13: Literature Review Snapshot

Speaker: Harshith | *On screen: 20 core papers plus 10 real-dataset cross-check papers*

- Our literature review has two parts. The core review covers 20 papers across 5 categories, with 3 reviewed in depth, including Gopakumar et al. 2026, our base paper.
- Those 5 categories: conformal prediction foundations, Mondrian and conditional-coverage CP, surrogate modeling and uncertainty quantification, queueing theory and ED operations research, and discrete-event simulation with ED-specific machine learning.
- On top of that, we have 10 papers that each report real ED service-time or length-of-stay data from their own hospital datasets - we use these to cross-check our own DES's service-time calibration numerically, not just cite them for general support.
- The review also includes a critical assessment section - specifically looking at where the reviewed literature's own assumptions do, or don't, transfer to a discrete-event queueing domain like ours.

Slide 14: Future Scope - End-Sem

Speaker: Harshith | *On screen: What's left in the remaining ~20%*

- To close: standard CP, Mondrian CP, and our core coverage-gap finding on Slide 11 are done. What's left is the remaining roughly 20 percent of the project.
- First, we'll stress-test exchangeability with an out-of-distribution demand-surge scenario, to see exactly where standard CP and Mondrian CP hold up, or break down, once that assumption is genuinely violated.

- Second, we'll finalize the full comparison - GP baseline versus standard CP versus Mondrian CP - on both coverage and interval width, across every target, not just the headline one.
- And finally, we'll bring it all together in the end-sem report and presentation: a fully quantified, written-up answer on whether Mondrian CP closes the marginal-versus-conditional coverage gap in this domain. Thank you.